

APPENDICES FOR REVIEWERS

APPENDIX A: ROBUSTNESS CHECKS

Our main results support our stated hypotheses regarding user and moderator bot activity (Table 1 below, replicated from Table 6 in the manuscript). In our main analysis, H1 was supported ($\beta_5 = -.679^{**}$), H2 was supported ($\beta_5 = .071^*$), and H3 was supported ($\beta_6 = -.153^{***}$). This indicates that the reflexive activity of user bots is associated with less human direct reciprocity (H1), user bot activity is associated with higher generalized reciprocity (H2), and moderator bot activity is negatively associated with preferential attachment as measured by decreased in-degree centralization (H3).

Table 1: PVAR results (Robust standard errors in parentheses, * $p < 0.05$, ** $p < 0.01$, * $p < 0.001$)**

$DV_{s,m} =$	Direct reciprocity _{s,m}	Generalized reciprocity _{s,m} (measured with density)	Preferential attachment _{s,m}
$\beta_1 DV_{s,m-1}$	.097*** (.027)	.643*** (.032)	.234*** (.029)
β_2 Community activity _{s,m-1}	.000* (.000)	-.000 (.000)	.000* (.000)
β_3 Community moderation _{s,m-1}	-.000 (.001)	.000* (.000)	.000 (.000)
β_4 Community membership _{s,m-1}	-.000 (.000)	-.000 (.000)	-.000** (.000)
β_5 User bot activity _{s,m-1}	-.679** (.258)	.071* (.034)	-.080 (.110)
β_6 Moderator bot activity _{s,m-1}	-.064 (.035)	-.003 (.006)	-.153*** (.023)

In this appendix, we describe the robustness checks we performed and the results as they compare to our main model specification. For each robustness check below, we describe why the check is needed, discuss how we modified our analyses, and present and analyze the results. We then discuss how the checks are consistent/inconsistent with our main findings. Each check in this discussion is named consistently with Figures A.1-A.3. Figures A.1-A.3 also include our main results for comparison (**Baseline**). Finally, we include a summary table (Table A.11) aggregating the results of our checks.

A.1 MEASUREMENT VALIDATION

To ensure that our empirical modeling is robust to our operationalization decisions, we conducted a number of robustness checks regarding our specific choices for independent variables in our model.

A.1.1 Changing Network Dynamics

One of the control variables in our main analysis is the total number of comments in a subreddit-month. While this covers changes in who is posting (more users may lead to more posts), it does not explicitly account for changes in human-human network composition over time. Therefore, we conduct a robustness check that includes the month-to-month change in the number of active human users in the subreddit as a control variable. For all hypotheses, our results are consistent (Figures A1-A3, Changing Network Dynamics). In this analysis (Table 2), H1 was supported ($\beta_5 = -.381^{**}$), H2 was supported ($\beta_5 = .174^*$), and H3 was supported ($\beta_6 = -.427^{***}$). This indicates that the reflexive activity of user bots is associated with less human direct reciprocity (H1), user bot activity is associated with higher generalized reciprocity (H2), and moderator bot activity is negatively associated with preferential attachment as measured by decreased in-degree centralization (H3).

Table 2: PVAR results with network dynamics included (* $p < 0.05$, ** $p < 0.01$, * $p < 0.001$)**

$DV_{s,m} =$	Direct reciprocity $_{s,m}$	Generalized reciprocity $_{s,m}$	Preferential attachment $_{s,m}$
$\beta_1 DV_{s,m-1}$	.047**	.426***	.195***
β_2 Community activity $_{s,m-1}$	.000	-.000	-.000
β_3 Community moderation $_{s,m-1}$	-.001**	.001**	.001
β_4 Change in community membership $_{s,m-1}$	.000	.000	-.000
β_5 User bots' activity $_{s,m-1}$	-.381**	.174*	.418
β_6 Moderator bot activity $_{s,m-1}$	-.139***	-.010	-.427***

A.1.2 Small accounting for false negatives

We explicitly account for potential false negatives (i.e., bots that are in the human-human interaction network and not included in our bot list) by randomly removing a percentage of humans from the network. The intuition here is to simulate the removal of bots that are not included in our list and therefore considered humans in the interaction network. For this robustness check, we conservatively assume that the number of bots on Reddit is double the number of bots on our original list. We calculated the percentage of users in our dataset that are bots (0.003%) and randomly removed the same percentage of human users from our network. For H1 and H2, our results are consistent (Figures A1 and A2, Small accounting for false negatives). In this analysis (Table 3), H1 was supported ($\beta_5 = -.679^{**}$), H2 was supported ($\beta_5 = .070^*$), and H3 was supported ($\beta_6 = -.159^{***}$). This indicates that the reflexive activity of user bots is associated with less human direct reciprocity (H1), user bot activity is associated with higher generalized reciprocity (H2), and moderator bot activity is associated with preferential attachment as measured by in-degree centralization (H3).

Table 3: PVAR results with a small adjustment in network to account for false negatives
(* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$)

DV _{s,m} =	Direct reciprocity _{s,m}	Generalized reciprocity _{s,m}	Preferential attachment _{s,m}
β_1 DV _{s,m-1}	.095***	.655***	.222***
β_2 Community activity _{s,m-1}	.000*	-.000	.000
β_3 Community moderation _{s,m-1}	-.000	.000	.000
β_4 Community membership _{s,m-1}	-.000	-.000	-.000**
β_5 User bots' activity _{s,m-1}	-.679**	.070*	-.049
β_6 Moderator bot activity _{s,m-1}	-.066	-.000	-.159***

A.1.3 Moderate accounting for false negatives

We make an even more conservative accounting for potential false negatives by randomly removing a larger percentage of humans from the network. Here we increase our assumption

about the user bot activity by a factor of 10 and randomly drop 0.03% of the human users from the interaction networks.

For this check, our results are consistent (Figures A1 and A2, Moderate accounting for false negatives). In this analysis (Table 4), H1 was supported ($\beta_5 = -.678^{**}$), H2 was supported ($\beta_5 = .069^*$), and H3 was supported ($\beta_6 = -.159^{***}$). This indicates that the reflexive activity of user bots is associated with less human direct reciprocity (H1), user bot activity is associated with higher generalized reciprocity (H2), and moderator bot activity is associated with preferential attachment as measured by in-degree centralization (H3).

Table 4: PVAR results with a larger adjustment in network to account for false negatives (* $p < 0.05$, ** $p < 0.01$, * $p < 0.001$)**

DV _{s,m} =	Direct reciprocity _{s,m}	Generalized reciprocity _{s,m}	Preferential attachment _{s,m}
β_1 DV _{s,m-1}	.095***	.655***	.222***
β_2 Community activity _{s,m-1}	.000*	-.000	.000
β_3 Community moderation _{s,m-1}	-.000	.000	.000
β_4 Community membership _{s,m-1}	-.000	-.000	-.000**
β_5 User bots' activity _{s,m-1}	-.678**	.069*	-.048
β_6 Moderator bot activity _{s,m-1}	-.066	-.000	-.159***

A.1.4 No human moderators in the network

It may be the case that the (human) moderators of a given subreddit drive most of the activity and significantly affect the network structure of the community (e.g., by posting to enforce norms). To test whether the human moderators affect our results, we remove all human moderators for a given subreddit from the human-human interaction network for that subreddit. With this check, we want to look exclusively at participatory interaction in the human-human interaction network and not supervisory interaction. We obtained the list of subreddit moderators using the Reddit API and excluded all moderators from our human-human interaction network.

We keep community moderation as a dependent variable (β_3 Community moderation) to control for the presence of moderation independent of moderator activity.

For this check, our results are consistent (Figures A1 and A2, No Human Moderators). In this analysis (Table 5), H1 was supported ($\beta_5 = -.519^{**}$), H2 was supported ($\beta_5 = .093^{**}$), and H3 was supported ($\beta_6 = -.157^{***}$). This indicates that the reflexive activity of user bots is associated with less human direct reciprocity (H1), user bot activity is associated with higher generalized reciprocity (H2), and moderator bot activity is associated with preferential attachment as measured by in-degree centralization (H3).

Table 5: PVAR results when excluding human moderators from the interaction network
(* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$)

DV _{s,m} =	Direct reciprocity _{s,m}	Generalized reciprocity _{s,m}	Preferential attachment _{s,m}
β_1 DV _{s,m-1}	.111***	.654***	.259***
β_2 Community activity _{s,m-1}	.000*	-.000	.000*
β_3 Community moderation _{s,m-1}	.000	-.000	.001*
β_4 Community membership _{s,m-1}	.000	-.000	-.000**
β_5 User bots' activity _{s,m-1}	-.519**	.093**	.001
β_6 Moderator bot activity _{s,m-1}	-.071**	.006	-.157***

A.1.5 Pastebin bot list

The list of bots we use for our main analysis is not an official list from Reddit. In fact, there is no official accounting of bots on Reddit because the bots look just like users on the platform, except that their activity is programmatically generated. To examine the effect of our bot list on our analyses, we replaced our list of bots with a new list. This new list contains 930 bots, which is shorter than our main list, even with a smaller threshold considered.¹ This list is posted online

¹ This list was obtained from the web, and the web page has since been deleted.

and does not have any details on how it was collected. Therefore it should not be taken as a comprehensive or authoritative list of bots.

In this analysis (Table 6), H1 was significant in the opposite direction ($\beta_5 = 1.098^*$), H2 was not supported ($\beta_5 = -.105$ n.s.), and H3 was not supported ($\beta_6 = -.294$ n.s.). This indicates that the reflexive activity of user and moderator bots is associated with more human direct reciprocity (H1), user and moderator bot activity is not associated with higher generalized reciprocity (H2), and moderator bot activity is not significantly associated with preferential attachment as measured by decreased in-degree centralization (H3).

Table 6: PVAR results when using an alternative list of bots (* $p < 0.05$, ** $p < 0.01$, * $p < 0.001$)**

DV _{s,m} =	Direct reciprocity _{s,m}	Generalized reciprocity _{s,m}	Preferential attachment _{s,m}
β_1 DV _{s,m-1}	.197***	.708***	.289***
β_2 Community activity _{s,m-1}	.000	.000	.000***
β_3 Community moderation _{s,m-1}	-.001	.000	-.000
β_4 Community membership _{s,m-1}	.000	-.000***	-.000***
β_5 User bots' activity _{s,m-1}	1.098*	-.105	-.084
β_6 Moderator bot activity _{s,m-1}	2.178*	-.215	-.294

The purpose of this robustness check was to analyze the influence of bots on Reddit using a different list of bots than the one in our main analysis. However, this list has a number of issues, which we believe contribute directly to the poor results shown in Table 6. First, the list only contains 930 bots, which is fewer than even our reduced consideration list (A.1.7 below). Second, the list was posted anonymously online, with no details on how it was collected or validated. This is consistent with much of the information online regarding Reddit bots. Collecting and classifying bots is at the moment neither comprehensive or definitive. While we leverage a semi-automated process for identifying the bots in our data (see Appendix C below), this indicates a need in future work for bot taxonomies, possibly hosted by the platform itself.

While there are several other curated lists of bots (e.g., on Reddit pages), none are as comprehensive as the data-driven list that we use in our main analyses. We performed a manual inspection of several such lists, including the list used in this robustness check, and found that they were subsets of our original list, further validating the data-driven approach to bot selection utilized in our main analyses.

A.1.6 Increased bot threshold

Our main experiment considers the top 2000 bots from our Reddit bot list. However, it may be the case that certain bots are excluded from our analyses at this cutoff (i.e., there are more than 2000 active bots on Reddit). Therefore, we set the bot threshold equal to 3000 instead of 2000, looking at the top 3000 bots in our bot list. With this check, we hope to confirm that our hypotheses are robust to the list of bots included in our analyses. In this case, we note that this potentially increases the number of false positives (humans included as bots in the list), which may reduce the amount of human activity in the network.

In this analysis (Table 7), H1 was supported ($\beta_5 = -.851^{**}$), H2 was supported ($\beta_5 = .075^*$), and H3 was supported ($\beta_6 = -.162^{***}$). This indicates that the reflexive activity of user bots is associated with less human direct reciprocity (H1), user bot activity is associated with higher generalized reciprocity (H2), and moderator bot activity is associated with lower preferential attachment as measured by increased in-degree centralization (H3).

Table 7: PVAR results when increasing the bot threshold (* $p < 0.05$, ** $p < 0.01$, * $p < 0.001$)**

DV _{s,m} =	Direct reciprocity _{s,m}	Generalized reciprocity _{s,m}	Preferential attachment _{s,m}
β_1 DV _{s,m-1}	.094***	.655***	.225***
β_2 Community activity _{s,m-1}	.000*	-.000	.000
β_3 Community moderation _{s,m-1}	-.000	.000	.000
β_4 Community membership _{s,m-1}	-.000	-.000	-.000**
β_5 User bots' activity _{s,m-1}	-.851**	.075*	.008
β_6 Moderator bot activity _{s,m-1}	-.042	-.001	-.162***

A.1.7 Decreased bot threshold

In the previous section, we were concerned with a short bot list that did not account for all of the bots on Reddit. Here we are interested in the opposite case: perhaps we include humans in our bot list and exclude them from the interaction network. Therefore, we set the bot threshold equal to 1000 instead of 2000, looking at the top 1000 bots in our bot list. Our goal is again to confirm that our results are robust to the list of bots included in the analyses. In contrast with the previous check, this check potentially increases the number of false negatives (bots that are missed and therefore considered humans), which may artificially inflate the number of human comments and reduce user bots' activity on a given network.

In this analysis (Table 8), H1 was supported ($\beta_5 = -1.252^{**}$), H2 was supported ($\beta_5 = .270^{***}$), and H3 was not supported ($\beta_6 = -.072$ n.s.). This indicates that the reflexive activity of user bots is associated with less human direct reciprocity (H1), user bot activity is associated with higher generalized reciprocity (H2), and moderator bot activity is not associated with a change in preferential attachment as measured by increased in-degree centralization (H3).

Table 8: PVAR results when decreasing the bot threshold (* $p < 0.05$, ** $p < 0.01$, * $p < 0.001$)**

DV _{s,m} =	Direct reciprocity _{s,m}	Generalized reciprocity _{s,m}	Preferential attachment _{s,m}
β_1 DV _{s,m-1}	.109***	.709***	.317***
β_2 Community activity _{s,m-1}	.000	-.000	.000*
β_3 Community moderation _{s,m-1}	-.000	-.001***	.000
β_4 Community membership _{s,m-1}	.000	.000	-.000**
β_5 User bots' activity _{s,m-1}	-1.252**	.270***	.067
β_6 Moderator bot activity _{s,m-1}	-1.318***	.288***	-.072

There is a very specific reason for H3 no longer being supported: the *AutoModerator* bot is ranked 1849 in our list. Therefore, when we do not include it as a bot, it is included in our

human-human interaction network. The fact that its inclusion in the human-human network leads to H3 no longer being supported is strong counterfactual evidence for H3.

A.1.8 Strict inclusion criteria for bot activity

We reduce the number of subreddits in our sample from 1074 to 145. To do this, we narrow the inclusion criteria in terms of the percentage of activity that is bots in the network. In our main analysis, we consider subreddits where the percentage of bot activity is between 3% and 30% of the total activity over the period. For our strict inclusion criteria, we only consider subreddits where the percentage of bot activity is between 10% and 20% of the total activity over the period.

In this analysis (Table 9), H1 was supported ($\beta_5 = -1.704^{***}$), H2 was not supported ($\beta_5 = -.013$ n.s.), and H3 was supported ($\beta_6 = -.093^{**}$). This indicates that the reflexive activity of user bots is associated with less human direct reciprocity (H1), user bot activity is not associated with higher generalized reciprocity (H2), and moderator bot activity is associated with lower preferential attachment as measured by increased in-degree centralization (H3).

Table 9: PVAR results with strict bot inclusion criteria (* $p < 0.05$, ** $p < 0.01$, * $p < 0.001$)**

$DV_{s,m}$	Direct reciprocity _{s,m}	Generalized reciprocity _{s,m}	Preferential attachment _{s,m}
$\beta_1 DV_{s,m-1}$	.146***	.505***	.341***
β_2 Community activity _{s,m-1}	.000***	-.000**	-.000
β_3 Community moderation _{s,m-1}	-.001	.002**	-.003**
β_4 Community membership _{s,m-1}	-.000**	.000	.000
β_5 User bots' activity _{s,m-1}	-1.704***	-.013	.927***
β_6 Moderator bot activity _{s,m-1}	.021	-.021	-.093**

By placing a severe restriction on the subreddit communities under consideration, we exclude those with relatively high and relatively low percentages of bot activity. Therefore, it makes sense that the generalized exchange in these networks would be less affected by bot activity, specifically with regard to those subreddits with higher bot activity that are included in our main

analyses but excluded in this check. However, we do see that H1 and H3 are still supported in this check, giving support to the idea that bots are impacting human-to-human interactions, and that there is a distinct difference in the effect of reflexive bots and supervisory bots.

A.1.9 Moderate inclusion criteria for bot activity

We reduce the number of subreddits in our sample from 1074 to 518. To do this, we narrow the inclusion criteria in terms of the percentage of activity that is bots in the network. In this analysis, we look at subreddits where the percentage of bot activity is between 5% and 25% of total activity over the period.

In this analysis (Table 10), H1 was supported ($\beta_5 = -1.393^{***}$), H2 was supported ($\beta_5 = .395^{***}$), and H3 was supported ($\beta_6 = -.103^{***}$). This indicates that the reflexive activity of user bots is associated with less human direct reciprocity (H1), user bot activity is associated with higher generalized reciprocity (H2), and moderator bot activity is associated with lower preferential attachment as measured by in-degree centralization (H3).

Table 10: PVAR results with moderate bot inclusion criteria (* p<0.05, ** p<0.01, * p<0.001)**

DV _{s,m} =	Direct reciprocity _{s,m}	Generalized reciprocity _{s,m}	Preferential attachment _{s,m}
β_1 DV _{s,m-1}	.113***	.751***	.310***
β_2 Community activity _{s,m-1}	.000*	-.000	.000
β_3 Community moderation _{s,m-1}	-.001**	.000	-.000
β_4 Community membership _{s,m-1}	.000	-.000	-.000**
β_5 User bots' activity _{s,m-1}	-1.393***	.395***	.019
β_6 Moderator bot activity _{s,m-1}	.028	-.030***	-.103***

A.1.10 Outlier Exclusion

To ensure that the results are not driven by the outliers in the data, we removed communities with very high values (1% outliers) of bot activity (i.e., user or moderator bot activity greater than 90% of the activity). This excludes 7 communities and 371 total observations (the new

sample contains 61,342 rather than 61,713 total observations). The results (below) are consistent with the main analysis.

Table 11: PVAR results with outliers excluded (* $p < 0.05$, ** $p < 0.01$, * $p < 0.001$)**

$DV_{s,m}$	Direct reciprocity $_{s,m}$	Generalized reciprocity $_{s,m}$	Preferential attachment $_{s,m}$
$\beta_1 DV_{s,m-1}$	.164**	.676***	.298***
β_2 Community activity $_{s,m-1}$	.000	-.000	.000*
β_3 Community moderation $_{s,m-1}$	-.001	.000	.000
β_4 Community membership $_{s,m-1}$	.000	-.000	-.000***
β_5 User bots' activity $_{s,m-1}$	-1.948**	.125*	.096
β_6 Moderator bot activity $_{s,m-1}$	.127	-.005	-.160***

A.2 ALTERNATIVE SPECIFICATIONS OF NETWORK PATTERNS

In our baseline experiment, we operationalized direct reciprocity as network reciprocity, generalized reciprocity as the network density, and preferential attachment as in-degree centralization (Section 3.2 in the manuscript). However, it is possible that other operationalizations of these dependent variables affect our analyses. Therefore, to test the robustness of our results, we change how we operationalize our three hypotheses.

A.2.1 Direct Reciprocity

We consider transitivity as our alternative operationalization for H1. Transitivity is defined as the tendency of nodes to cluster together (Block, 2015). Colloquially, “the friends of my friends are my friends.” However, in the context of our analyses, transitivity may not be the best metric for measuring direct reciprocity for several reasons. First, transitivity is typically considered for undirected graphs, while we operationalize the human-human interaction network as a directed network. Second, there is prior work in the literature showing that the relationship between transitivity and direct reciprocity is negative (Block, 2015). However, to inform future research,

we report on this specification. We note that transitivity is not significantly impacted by user bots' activity (Table 12).

A.2.2 Generalized Reciprocity

We consider flow hierarchy as an alternative operationalization of generalized reciprocity (Luo & Magee, 2011). Flow hierarchy is defined as the percentage of links that are not part of a cycle in a directed graph. Flow hierarchy typically represents the consistent directionality of links in a network (e.g., nodes in a supply chain). This is a stricter operationalization of indirect reciprocity than our main metric of (log) edges in the network. We are concerned with the flow of information and connections in the network generally, not necessarily in a strict flow direction. We note that flow hierarchy is significantly impacted by user bot activity (Table 12).

A.2.3 Preferential Attachment

As an alternative operationalization for preferential attachment, we look at out-centrality instead of in-centrality. Our results are consistent with our main model (Figure A3, Alternative Operationalization).

Table 12: PVAR results for alternative DV specifications (* p<0.05, ** p<0.01, * p<0.001)**

DV _{s,m} =	Transitivity _{s,m}	Flow hierarchy _{s,m}	Out-centrality _{s,m}
β_1 DV _{s,m-1}	.328***	.189***	.342***
β_2 Community activity _{s,m-1}	.000**	-.000	.000*
β_3 Community moderation _{s,m-1}	-.001***	.004***	.000
β_4 Community membership _{s,m-1}	-.000*	-.000***	-.000**
β_5 User bots' activity _{s,m-1}	-.107*	.234*	-.084
β_6 Moderator bot activity _{s,m-1}	.004	.175***	-.053***

To summarize the results of our check on measure specification (Table 11), H1 was supported ($\beta_5 = -.107^*$), H2 was supported ($\beta_5 = .234^*$), and H3 was supported ($\beta_6 = -.053^{***}$). This indicates that the reflexive activity of user bots is associated with less human direct reciprocity (H1), user bot activity is associated with higher generalized reciprocity (H2), and moderator bot activity is

negatively associated with preferential attachment as measured by decreased out-degree centralization (H3).

A.3 Summary of Robustness Checks

In Figures A.1 - A.3, we are concerned with two questions related to our checks: (1) is the sign of the estimated coefficient consistent with our main model specification, and (2) is the robustness check coefficient significant at the $p < 0.05$ level? Overall, our robustness check results are consistent with our main model specification (i.e., mostly triangles in Figures A1-A3 below).

The robustness checks show that our findings are generalizable to a larger sample, but communities indeed differ in terms of their propensity to be affected by bots.

When replacing our bot list with a different list (A.1.5) we find that our results are not significant. We believe that this list is a poor one, especially because it is no longer available on the web page that we retrieved it from. For comprehensiveness, we include this check in our results.

When we reduce the number of bots under consideration from 2000 to 1000 (A.1.7), we find that H3 is no longer supported, while H1 and H2 are supported. We believe this is due to the fact that the AutoModerator bot is ranked 1849 in our bot listing. Therefore, when we reduce to 1000 bots, AutoModerator is no longer considered a bot but is included in the human-human interaction network. The inclusion of this bot specifically affects H3 because it is the main supervisory bot on the Reddit platform (noting of course that its behavior can be customized for each subreddit that chooses to deploy it). By excluding AutoModerator from our bot analysis we are excluding the main mechanism for affecting preferential attachment (H3).

When we strictly reduce the number of subreddits under consideration (A.1.8), we see that H2 is no longer supported. The network density changes are not present, and we believe this is due to the fact that those networks with more bot activity will also see more of a change in density. This assumption is confirmed when we relax the inclusion criteria to a more moderate restriction from our main analysis (A.1.9).

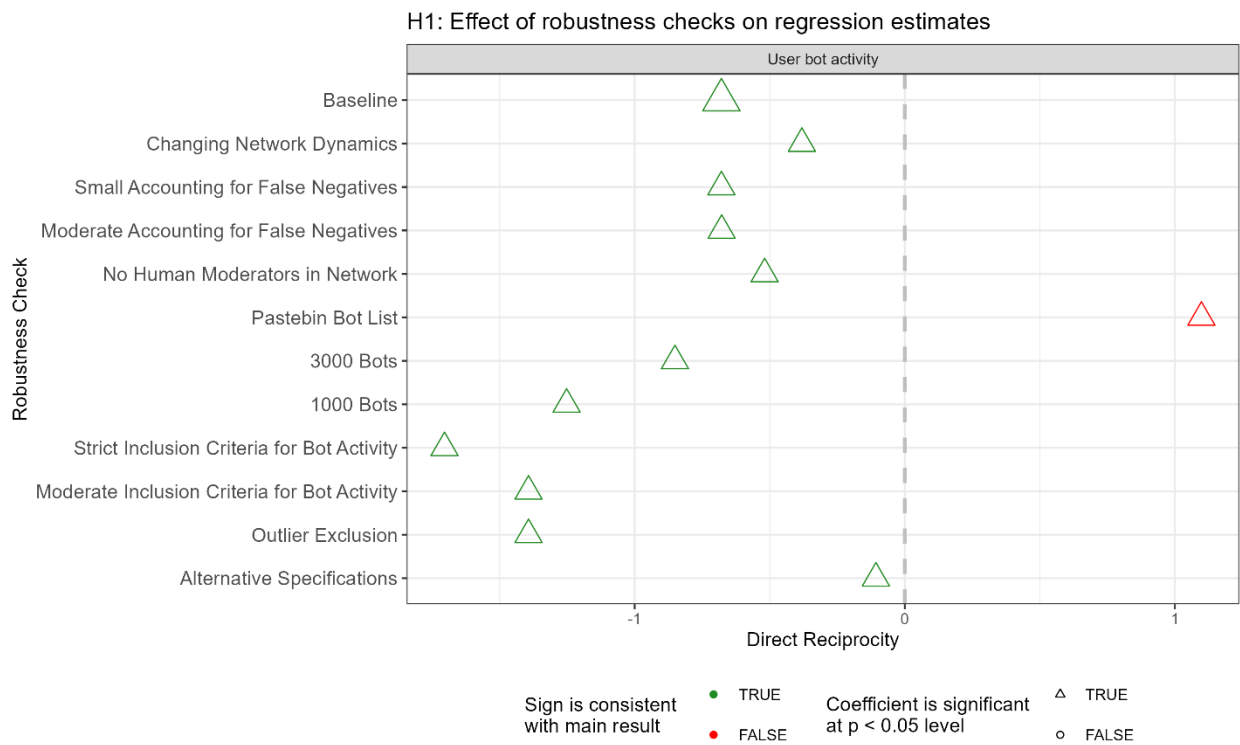


Figure A1: Results of robustness checks for H1.

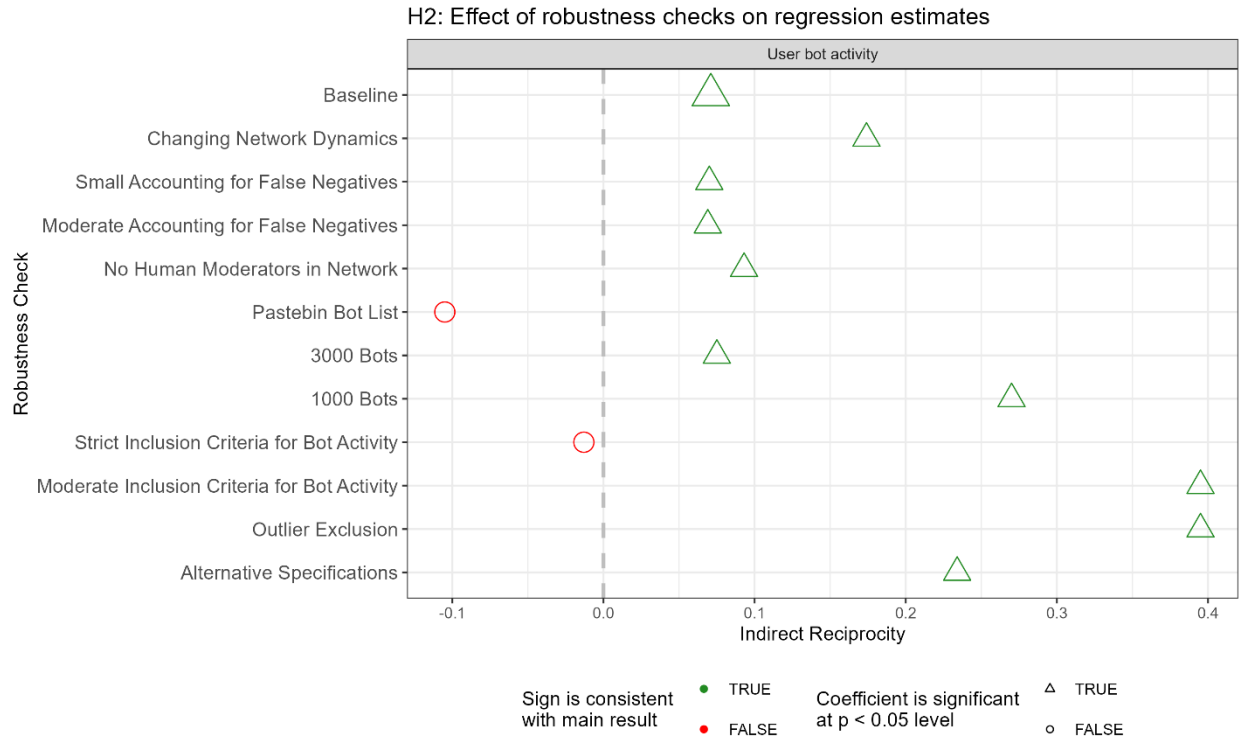


Figure A2. Results of robustness checks for H2.



Figure A3. Results of robustness checks for H3.

Table 13. Summary of robustness check results

Analysis	Hypothesis 1	Hypothesis 2	Hypothesis 3
Main	Negative, significant	Positive, significant	Negative, significant
<i>Robustness Tests in Appendix A</i>			
A.1.1	Negative, significant	Positive, significant	Negative, significant
A.1.2	Negative, significant	Positive, significant	Negative, significant
A.1.3	Negative, significant	Positive, significant	Negative, significant
A.1.4	Negative, significant	Positive, significant	Negative, significant
A.1.5	Positive, significant	Negative, not significant	Negative, not significant
A.1.6	Negative, significant	Positive, significant	Negative, significant
A.1.7	Negative, significant	Positive, significant	Negative, not significant
A.1.8	Negative, significant	Negative, not significant	Negative, significant
A.1.9	Negative, significant	Positive, significant	Negative, significant
A.1.10	Negative, significant	Positive, significant	Negative, significant
A.2	Negative, significant	Positive, significant	Negative, significant

APPENDIX B

We outlined below the panel VAR-Granger causality Wald test (Abrigo & Love, 2016). In this test, H_0 : Excluded variable does not Granger-cause Equation variable, and H_a : Excluded variable Granger-causes Equation variable.

Equation \ Excluded	χ^2	df	Prob > χ^2
Direct reciprocity			
Community activity	0.628	1	0.428
Community moderation	3.318	1	0.069
Community density	140.233	1	0.000
User bots' activity	22.015	1	0.000
Moderator bot activity	44.133	1	0.000
ALL	390.380	5	0.000
Generalized reciprocity			
Community activity	11.514	1	0.001
Community moderation	10.156	1	0.001
Community density	14.190	1	0.000
User bots' activity	4.210	1	0.040
Moderator bot activity	1.990	1	0.158
ALL	31.755	5	0.001
Preferential attachment			
Community activity	2.389	1	0.122
Community moderation	1.394	1	0.238
Community density	18.373	1	0.000
User bots' activity	0.859	1	0.354
Moderator bot activity	59.861	1	0.000
ALL	74.878	5	0.000

APPENDIX C

In this Appendix, we discuss our bot identification strategy. On Reddit, as in many other social media contexts, accounts that are bots do not need to be explicitly revealed as such. They are user accounts that act programmatically. However, bot programmers often indicate that a particular account is a bot to collect feedback for the bot's performance. Moreover, the Reddit community has adopted a convention around sharing feedback for bots. We leverage these two pieces of information to identify bot accounts on Reddit for our analysis.

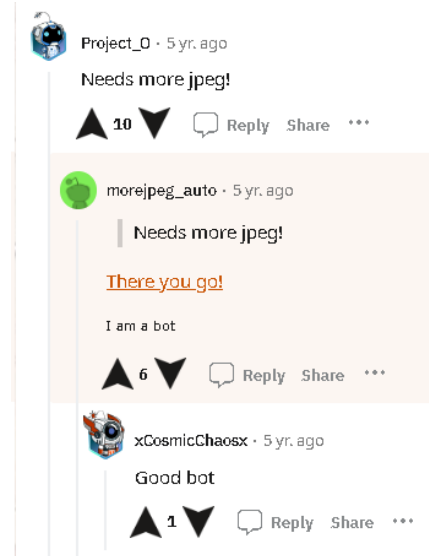


Figure C1. Example of a human user indicating approval of bot performance with a “good bot” reply. Source:
<https://www.reddit.com/r/ElderScrolls/comments/85n5bq/comment/dvytqsb/?context=8&depth=9>

On Reddit, when a bot posts, users often reply with “good bot” or “bad bot” to signify that they like or dislike the content, respectively (Figure C1). Figure C1 shows an example of this behavior. The *morejpeg_auto* bot is activated when a user posts “more jpeg” or a similar trigger phrase. The bot then crawls the comment thread for a linked image and attempts to enhance the image. It then posts a new link to the updated image. After this, one or more (human) accounts reply to the bot’s comment with “good bot” or “bad bot” to signify approval or disapproval of the activity, respectively.

In an attempt to both identify and rank bots on Reddit, individuals have scraped this reply behavior to score bots based on the number of good bot and bad bot votes that they receive. We leveraged the Botrank website for data collection, which scraped Reddit data to track accounts that received many “good bot” or “bad bot” replies to generate a bot ranking list.²

² In the past, members of the Reddit community have tried to “game” this ranking system by inflating the number of good bot (or bad bot) replies for certain bots. While the specific ranking is not critical to our analyses, we can use

The botrank list is likely to contain false positives, that is, human accounts listed as bots. This is due to the fact that the ranking looks for “good bot” and “bad bot” replies, which can be posted to comments written by humans. To confirm the validity of our bot account listing, we designed a semi-automatic labeling procedure.

For each account on the list, we check the following items:

1. Does the account have the word “bot” in its username (Figure C2)?
2. Does the account list that it is a bot in its profile (Figure C2)?
3. Does the account indicate that it is a bot in the footer of its posts (Figure C1)?
4. Does the account link to a GitHub page (where its code is stored) in its profile (Figure C2)?
5. Does the account link to a GitHub page (where its code is stored) in its comment footer (Figure C3)?

Each of the above can be checked programmatically. For the username check (#1 above), we used string matching. For the other items (#2-5), we used the Reddit API to check the relevant information for each account.



Figure C2. The account *GifReversingBot* meets criteria 1, 2, and 4 above, based on an inspection of the account profile page.

this list to reduce the number of possible bots from the millions of active Reddit users to a more manageable number.

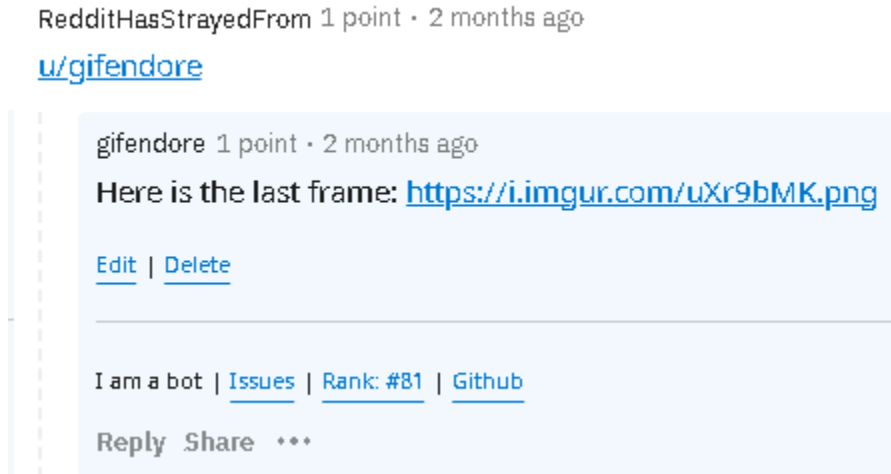


Figure C3. The *gifendore* bot links to its source code at the end of each post, satisfying criteria 5 above.

We ran this check for the first 3000 accounts on the list.

Table C1. Results of the automated checks.

“Bot” in			“GitHub” in		
Name	Profile Description	Comment	Description	Comment	One or More Hit
1390	251	764	35	209	1648

After this step, for those accounts ranked 1-2000 that did not meet one of the above criteria (N=618), we manually inspected each account. At this stage, we checked for the following:

1. Does the account post based on programmatic triggers?
2. Is the content varied enough that it is not clearly the behavior of a bot?
3. Does the account self-identify as a human?

This manual inspection added 146 bots to our analysis. We note here that between ranks 2001 and 3000, 734 accounts do not meet one or more of our automated criteria.

While this categorization is imperfect, we believe that it will significantly reduce the number of false positives (humans identified as bots) in our analyses. In our robustness checks, our results

are consistent when we both reduce and expand our bot list to identified bots ranked between 1-1000 and 1-3000, respectively (Refer to **A.1.6** and **A.1.7**).

REFERENCES TO THE APPENDIX

- Abrigo, M. R. M., & Love, I. (2016). Estimation of panel vector autoregression in Stata. *Stata Journal*, 16(3), 778–804. <https://doi.org/10.1177/1536867x1601600314>
- Block, P. (2015). Reciprocity, transitivity, and the mysterious three-cycle. *Social Networks*, 40, 163–173. <https://doi.org/https://doi.org/10.1016/j.socnet.2014.10.005>
- Costenbader, E., & Valente, T. W. (2003). The stability of centrality measures when networks are sampled. *Social Networks*, 25(4), 283–307.
- Luo, J., & Magee, C. L. (2011). Detecting evolving patterns of self-organizing networks by flow hierarchy measurement. *Complexity*, 16(6), 53–61.